

Homework 2

Due: Friday, February 18

All homeworks are due at 11:59 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

LaTeX tips

As you begin using LaTeX, here are some useful commands that might come in handy for this homework:

<code>\Pow</code>	$\mathcal{P}$	<code>\cap</code>	$\cap$	<code>\cup</code>	$\cup$
<code>\star</code>	$\star$	<code>\overline{A}</code>	$\overline{A}$	<code>\emptyset</code>	$\emptyset$
<code>\times</code>	$\times$	<code>\setminus</code>	$\setminus$	<code>\subseteq</code>	$\subseteq$

A note that LaTeX is *not* required until **Homework 3**.

Problem 1

- Prove (using the “set element” method) or disprove the following claim:
For arbitrary sets A and B , $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.
- Prove (using the “set element” method) or disprove the following claim:
For arbitrary sets A and B , $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Solution:

Part 1: To disprove our first claim, it suffices to provide a counterexample. Let A be $\{1\}$. Let B be $\{3\}$.

$$\mathcal{P}(A \cup B) = \{\{1\}, \{3\}, \{1, 3\}, \emptyset\}.$$

$$\mathcal{P}(A) = \{\{1\}, \emptyset\}.$$

$$\mathcal{P}(B) = \{\{3\}, \emptyset\}.$$

Therefore, $\mathcal{P}(A) \cup \mathcal{P}(B) = \{\{1\}, \{3\}, \emptyset\}$, which is not $\{\{1\}, \{3\}, \{1, 3\}, \emptyset\}$. Thus $\mathcal{P}(A \cup B)$ is not equal to $\mathcal{P}(A) \cup \mathcal{P}(B)$.

Part 2:

Claim: For all sets A and B , $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Proof:

We will use the set element method to prove our claim. We will first show $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$, and we will then show $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$.

Consider some x in $\mathcal{P}(A \cap B)$. If x is in $\mathcal{P}(A \cap B)$, then x must be a subset of $A \cap B$. By the definition of intersection, x is a subset of A and x is a subset of B . This means x must be a member of $\mathcal{P}(A)$ and x must be a member of $\mathcal{P}(B)$. Therefore, x must be a member of $\mathcal{P}(A) \cap \mathcal{P}(B)$. Thus, $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$.

Now, consider some y in $\mathcal{P}(A) \cap \mathcal{P}(B)$. By the definition of intersection, y is in both $\mathcal{P}(A)$ and $\mathcal{P}(B)$. If y is in both $\mathcal{P}(A)$ and $\mathcal{P}(B)$, then y is a subset of both A and B . Thus, $\mathcal{P}(A \cap B)$ must also contain y . Thus, $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$.

Because $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$ and $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$, $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Rubric:

- a. +1 pt for providing counterexample sets
+1 pt for computing $\mathcal{P}(A \cup B)$ and $\mathcal{P}(A) \cup \mathcal{P}(B)$ and showing they are not equal
- b. +1 pt for using the set-element method
+1 pt for correctly showing $\mathcal{P}(A \cup B) \subset \mathcal{P}(A) \cup \mathcal{P}(B)$
+1 pt for correctly showing $\mathcal{P}(A) \cup \mathcal{P}(B) \subset \mathcal{P}(A \cup B)$

Problem 2

- a. A binary operation, $\star$, is an operation on a set that takes in two elements from the set and returns a third.

We say that a binary operation $\star$ on a set S is commutative if

$$x \star y = y \star x \text{ for all } x, y \in S.$$

Example

For example, the addition operation (+) and the subtraction operation (−) are binary operations on the integers. Addition is *commutative* while subtraction is not.

Let X be a finite set. For each of the following operations, prove whether or not the given operation is commutative over $\mathcal{P}(X)$.

- i. Set union
 - ii. Set intersection
 - iii. Set difference
 - iv. Symmetric difference¹
- b. Consider a binary operation $\star$ on a set S . An *identity element* for $\star$ is any $e \in S$ such that $e \star x = x \star e = x$ for all $x \in S$.

Example

For example, 0 is an identity element for the operation + over the integers, because $x + 0 = 0 + x = x$ for all $x \in \mathbb{Z}$.

Let X be a finite set. Which elements in $\mathcal{P}(X)$ are identity elements for the set union operation ($\cup$)? Which elements in $\mathcal{P}(X)$ are identity elements for the set intersection operation ($\cap$)? Prove your response.

Solution:

Part a:

¹Recall that the symmetric difference is defined as $(A \cup B) \setminus (A \cap B)$. See the lecture notes and/or text for definitions of any other operations.

- i. We want to show that $A \cup B = B \cup A$. We'll prove this via the set element method.

If $x \in A \cup B$, then $x \in A$ or $x \in B$. By definition, that also means $x \in B \cup A$. So any element in the RHS is also in the LHS and $A \cup B \subseteq B \cup A$.

Similarly, if $x \in B \cup A$, then $x \in B$ or $x \in A$, which makes $x \in A \cup B$. The LHS is a subset of the RHS and $B \cup A \subseteq A \cup B$.

$$A \cup B \subseteq B \cup A \text{ and } B \cup A \subseteq A \cup B \rightarrow A \cup B = B \cup A.$$

- ii. Set intersections are commutative, and the proof is identical to the one for the set union.
- iii. Set differences are *not* commutative. Consider $A = \{1, 2, 3, 4\}$ and $B = \{1\}$. Let $X = A \cup B$. Now, we know $P(X)$ contains A and B , so we will use these elements in our set difference. $A \setminus B = \{2, 3, 4\}$ but $B \setminus A = \emptyset$. Just as, in general, $a - b \neq b - a$, so too, in general, $A \setminus B \neq B \setminus A$.
- iv. We want to show that $(A \cup B) \setminus (A \cap B) = (B \cup A) \setminus (B \cap A)$. We already showed earlier that $A \cup B = B \cup A$ and $A \cap B = B \cap A$. So the left- and right-hand sides are exactly the same, and the symmetric difference is commutative.

Part b:

By the rules of powersets, we know that $\mathcal{P}(S)$ contains $\emptyset$ and S itself. Consider an arbitrary $x \in \mathcal{P}(S)$.

Union:

For the $\cup$ operation, an identity element is $\emptyset$. This is because the union $x \cup \emptyset = x$.

To show that no other e is the identity element, assume some x which is non-empty is the identity. Now, consider $X \cup \emptyset$. Because X is the identity, our expression should result in $\emptyset$.

Now, if X is non-empty, it must contain at least one element, say a .

$\{a, \dots\} \cup \emptyset = \{a, \dots\} \neq \emptyset$, and so X cannot be the identity. Thus, the empty set is the only identity.

Solution:

Intersection:

For the $\cap$ operation, an identity element is S itself. Because every single element in x is also in S , the intersection of the two sets is simply x . So S works as an identity element.

Assume for the sake of contradiction there is some other identity for intersection, and call it I . We know I must be some proper subset of S . Now, consider $S \cap I$. Because I is the identity, our expression should result in S . But if I is some proper subset of S , then $S \cap I$ must be some proper subset of S . Thus, the result of our intersection is lacking some element of S . Therefore, I cannot be the identity.

Note that for both $\cup$ and $\cap$, we need only consider $x \star e$, and not $e \star x$ as well, because we proved in part (a) that unions and intersections are commutative.

Rubric:

- a.
 - i. +2 pt for using set-element method to show union is commutative (1 pt for answer and 1 pt for proof)
 - ii. +2 pt for using set-element method to show intersection is commutative (1 pt for answer and 1 pt for proof)
 - iii. +2 pt for finding counterexample/otherwise logically proving set difference is not commutative (1 pt for answer and 1 pt for proof)
 - iv. +2 pt for using set-element method or using parts i. and ii. to show symmetric difference is commutative (1 pt for answer and 1 pt for proof)
- b. +1 pt for proving $\emptyset$ is the identity element of union
+1 pt for proving nothing else is an identity element of union
+1 pt for proving S is the identity element of intersection
+1 pt for proving nothing else is an identity element of intersection

Problem 3

a. Given the following sets:

$$A = \{a, b, c, d, g, f\}$$

$$B = \{a, e\}$$

$$C = \{b, d, e, g\}$$

$$D = \{a, g\}$$

- i. Compute $(A \cap C) \cup D$
 - ii. Compute $B \times ((A \cap C) \cup D)$
 - iii. Compute $|\mathcal{P}(B \times ((A \cap C) \cup D))|$
- b. Let A, B, C be arbitrary sets, and assume they are all subsets of a universal set U . Using **set algebra**, show that the following equivalences hold:

$$\text{i. } (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$$

$$\text{ii. } \overline{(A \cup (B \cap C))} = (\overline{C} \cup \overline{B}) \cap \overline{A}$$

$$\text{iii. } (A \cap B) \cup (B \cap C) \cup (C \cap A) = (B \cap (A \cup C)) \cup (C \cap (A \cup B))$$

Hint: Try gaining insight by going in reverse.

Solution:

- a.
 - i. $\{a, b, d, g\}$
 $(A \cap C) = \{b, d, g\}, D = \{a, g\}, (A \cap C) \cup D = \{a, b, d, g\}$
 - ii. $\{(a, a), (e, a), (a, b), (e, b), (a, d), (e, d), (a, g), (e, g)\}$
 $(A \cap C) = \{b, d, g\}, D = \{a, g\}, (A \cap C) \cup D = \{a, b, d, g\}$
 $B \times ((A \cap C) \cup D) = \{(a, a), (e, a), (a, b), (e, b), (a, d), (e, d), (a, g), (e, g)\}$
 - iii. 256
 $B \times ((A \cap C) \cup D) = \{(a, a), (e, a), (a, b), (e, b), (a, d), (e, d), (a, g), (e, g)\}$
 $|B \times ((A \cap C) \cup D)| = 8, 2^8 = 256$

b. i.

$$\begin{aligned} (A \setminus B) \cup (B \setminus A) &= (A \cap \overline{B}) \cup (B \cap \overline{A}) \\ &= (A \cup B) \cap (A \cup \overline{A}) \cap (\overline{B} \cup B) \cap (\overline{B} \cup \overline{A}) \\ &= (A \cup B) \cap \overline{(A \cap B)} \\ &= (A \cup B) \setminus (A \cap B). \end{aligned}$$

ii.

$$(A \cup (B \cap C))^C = A^C \cap (B \cap C)^C$$

$$\begin{aligned} &= A^C \cap (B^C \cup C^C) \\ &= (B^C \cup C^C) \cap A^C \\ &= (C^C \cup B^C) \cap A^C \end{aligned}$$

iii.

$$\begin{aligned} (A \cap B) \cup (B \cap C) \cup (C \cap A) &= (A \cap B) \cup (B \cap C) \cup (B \cap C) \cup (C \cap A) \\ &= (B \cap A) \cup (B \cap C) \cup (C \cap B) \cup (C \cap A) \\ &= (B \cap (A \cup C)) \cup (C \cap (A \cup B)) \end{aligned}$$

Rubric:

- a.
 - i. +1 pt for correct answer
 - ii. +1 pt for correct answer
 - iii. +1 pt for correct answer
- b.
 - i. +1 pt for correct answer, +1 pt for work
 - ii. +1 pt for correct answer, +1 pt for work
 - iii. +1 pt for correct answer, +1 pt for work

☕ Problem 4 (Mind Bender — *Extra Credit*)

Recall that we can create sets with *descriptions* using set-builder notation, like

$$A = \{x \in \mathbb{R} \mid x^2 > 1\}.$$

We would read this as “ A is the set of x in the real numbers *such that* $x^2 > 1$.” The condition that $x^2 > 1$ is a condition we place on this set to define it. Another example could be

$$H = \{h \mid h \text{ is a hot drink}\}.$$

to which $H = \{\text{hot coffee, hot tea, latte, } \dots\}$. This is an example of a description that is in natural language. One might consider if *every* description is a valid description.

Consider the description “ X does not contain itself”.

Example

For example, let

$$H = \{h \mid h \text{ is a hot drink}\}.$$

be the set of all hot drinks. This set H does not contain itself ($H \notin H$) since the set of all hot drinks is not a hot drink.

However, let

$$J = \{h \mid h \text{ is not a hot drink}\}.$$

be the set of everything that *isn't* a hot drink. Well, J is a set of things, that clearly isn't a hot drink, so $J \in J$ and we say J “contains itself”.

Let's build a set with this specific description. Let S be the set that contains all sets that do not “contain themselves”. That is, we define it to be

$$S = \{X \text{ is a set} \mid X \notin X\}$$

A set such as J *would not* be in S , as J contains itself. However, a set such as H *would* be in S , as H does not contain itself.

- Show that the assumption that S is a member of S leads to a contradiction.
- Show that the assumption that S is not a member of S leads to a contradiction.
- What do these contradictions suggest about how we can or cannot define a set? This paradox is called *Russell's Paradox*. Are there any ways to resolve these contradictions in set theory? Do some research and cite at least one source.²

²This is an open ended question, and there is no right answer! You should demonstrate to us that you have an understanding of what is going on.

Solution:

Part A: Assume that S is a member of S . If S is a member of S , then by the definition of S , S does not contain itself. But if S is a member of S , then S *does* contain itself. We therefore have that, if S is a member of S , S is not a member of S . That is, $S \in S \rightarrow S \notin S$.

Part B: Assume that S is not a member of S . If S is not a member of S , then S should be in S , as S contains all sets that do not contain themselves. We therefore have that if S is not a member of S , S is a member of S . That is, $S \notin S \rightarrow S \in S$.

Part C: (Answers will vary) This contradiction is called Russell's Paradox. Russell's paradox shows that the unrestricted axiom of comprehension (the idea that any condition or property may be used to determine a set) needs restriction because it can lead to contradictions. Russell's paradox is also motivated by adoption of the vicious circle principle (no propositional function can be defined prior to specifying the function's scope of application).

A mathematician named Ernst Zermelo then adapted a new form of Set Theory where the previous axiom of there existing a set containing any elements that satisfy a given property was replaced by an axiom that says that for any combination of set and property, there exists a new set of elements that are in the original set and that satisfy the property. This new axiomatic foundation avoids Russell's Paradox, and defines Zermelo-Frankel Set Theory.

Source: <https://www.scientificamerican.com/article/what-is-russells-paradox/>

Rubric:

- a. +0.5 for correct part A
- b. +0.5 for correct part B
- c. +1 for answering part c (≥ 2 sentences) in a way that shows learning and understanding and providing at least one source